Perception-Constrained Altitude Control for Aerial–Ground Cooperative Tracking with a Shared Downward Camera

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***Abstract*— We present a genuinely-controlled, reproducible software-in-the-loop (SITL) study of aerial–ground cooperative tracking in which a single downward UAV camera is the ground vehicle's *sole* exteroceptive sensor, serving simultaneously as its navigation reference (path segmentation) and its state estimator (ArUco/EKF). This shared-sensor architecture couples navigation and estimation through a single control variable — the UAV altitude — and this coupling is the organizing principle of our design: low altitude sharpens estimation (ground-sampling noise $R \propto z^2$) but starves navigation coverage (the path lookahead leaves the camera footprint), while high altitude does the reverse. Our contributions are threefold. (i) We formalize this coverage–resolution tension and derive a perception-constrained altitude policy (fly as low as segmentation allows, climb into turns), and we *demonstrate it closed-loop in SITL*: the policy actuates altitude over $[2.5, 5.1]$ m and dominates both fixed-altitude baselines — a fixed-low policy suffers coverage-loss transients ($9\times$ larger peak covariance as the target leaves the footprint in turns) while a fixed-high policy pays a 61% estimation penalty, and the adaptive policy attains the best of both. (ii) We show a perception-driven event trigger cuts the UAV nominal (aerial) planner's online re-solves $\approx 25\times$ ($797 \pm 16 \rightarrow 32 \pm 1$ per run) and its CPU $\approx 12\times$ at matched estimation quality (covariance 0.0095, $\approx 99.9\%$ visibility for both continuous and event-triggered aerial planning). (iii) We deliver an open, reproducible ROS 2/Gazebo testbed and establish a methodological prerequisite for closed-loop active-perception study: headless execution is required for reproducibility (≈ 2.7 m GUI vs. ≈ 0.03 m headless identical-configuration fork).**

I. INTRODUCTION

The deployment of Unmanned Ground Vehicles (UGVs) in unstructured and GPS-denied environments is heavily constrained by their limited perception horizons. Ground-level sensors such as LiDAR and stereo cameras suffer from occlusion, narrow fields of view (FoV), and an inability to anticipate upcoming path curvature. To mitigate this, aerial–ground cooperative systems employ an Unmanned Aerial Vehicle (UAV) as an elevated "eye in the sky" for a perception-limited UGV.

Most cooperative-tracking frameworks treat the UAV as a *passive* observer: it follows a predefined trajectory or simply minimizes distance to the UGV, treating the UGV's state estimation as a downstream byproduct [3]. This decoupling is fragile: when the UGV maneuvers aggressively or is occluded, the UAV can lose the target, the estimation covariance grows, and tracking degrades. Active-perception planning, where the sensor trajectory is optimized to maximize information gain [4], addresses part of this but typically relies on continuous, computationally expensive replanning and rarely integrates robust tracking guarantees for the aerial vehicle.

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This paper studies a *bidirectional* observability-aware architecture in which *both* vehicles adapt to the ground vehicle's estimation uncertainty. Its contributions, as realized in the accompanying open implementation, are:

- **A perception-constrained altitude policy, formalized and demonstrated closed-loop.** With one downward camera serving both the ground vehicle's navigation (path segmentation) and its state estimate (ArUco/EKF) [5], [6], low altitude sharpens estimation ($R \propto z^2$) but starves navigation coverage; the policy flies as low as segmentation allows and climbs into turns. In SITL it actuates altitude over $[2.5, 5.1]$ m and *dominates both fixed-altitude baselines*: it avoids the coverage-loss transients of a fixed-low policy and the estimation penalty of a fixed-high one (Sec. VII).
- **A genuinely-controlled, reproducible SITL realization** in ROS 2/Gazebo (vision/ArUco pipeline, view-quality-coupled EKF, belief-space UGV MPC, and the UAV planner/ancillary pair [1], [7]), including a methodological finding: headless execution is a prerequisite for reproducible closed-loop active-perception study (≈ 2.7 m vs. ≈ 0.03 m identical-configuration trajectory fork).
- **A validated co-design result**: the perception-driven event trigger cuts the UAV nominal (aerial) planner's online re-solves $\approx 25\times$ and its CPU $\approx 12\times$ at matched estimation quality, for a small loss of inter-trigger trajectory smoothness.
- **An open, reproducible ROS 2/Gazebo testbed** for closed-loop active-perception study, together with a methodological finding often left unstated: headless execution is a prerequisite for reproducibility (≈ 2.7 m vs. ≈ 0.03 m identical-configuration trajectory fork).

The shared camera makes navigation and estimation inseparable through altitude. We treat this coupling not as an artifact to be engineered away but as the central design constraint — it is precisely what makes the altitude policy necessary and what the event trigger exploits.

We build the online controller through a principled design pipeline: an offline robust-MPC synthesis (mRPI, constraint tightening) sizes the tube that the lightweight online adaptive law then tracks (Sec. VI), and we validate the computational claim in simulation. The remainder of the paper is organized as follows. Section II reviews related work; Section III states

the problem; Section IV gives the vehicle and perception models; Section V develops the coupled event-triggered tube-MPC; Section VI states the theoretical properties and the exact implementation status; and Section VII reports the simulation study.

II. RELATED WORK

Robust and tube MPC. Tube MPC [1], [8], [10] decomposes control into a nominal plan and an ancillary law that confines the true state to a robust positively invariant (RPI) tube; Raković et al. [2] give the standard outer approximation of the minimal RPI set, and the nonlinear case is treated in [9]. These guarantees assume a fixed, bounded disturbance set and constraint tightening computed offline. Time-varying (homothetic/parameterized) tubes generalize the fixed tube but require the scaling to enter the optimization with its own feasibility argument; our online adaptivity (Section V) is a lighter-weight surrogate.

Perception-aware and active planning. Perception-aware trajectory optimization for quadrotors [4], [14] and active visual localization [15] shape motion to serve estimation. Belief-space planning [5], [6], [16], [17] augments the planning state with the estimator covariance — closely related to active SLAM [19], [18]. Our belief-space UGV controller is in this family. Distinctively, our architecture is *bidirectional*: the *target* shapes its trajectory to stay observable while the *tracker* adapts both its altitude and its robustness margin to perception quality.

Cooperative aerial–ground teams. Using an aerial vehicle as an "eye in the sky" for a ground robot is well studied [22], including recent perception-limited-UGV navigation [20] and aerial-vision guidance of a perception-less ground vehicle [21], alongside quadrotor target-tracking under field-of-view and occlusion constraints [23], [24] and EKF-based cooperative localization [25], [26]. These systems typically fix the aerial vehicle's altitude and treat it as a passive relay; we show instead that altitude is an active perception control input whose optimal use is dictated by the shared-camera coupling, demonstrated in an integrated, reproducible SITL realization.

Event-triggered MPC. Event-triggered and self-triggered control [12], [11] re-solve only when a state-dependent condition fires, and extend to robust MPC of nonlinear systems [7], [13]. We instantiate a perception-driven trigger with a critical tube-exit condition plus anticipatory conditions.

III. PROBLEM FORMULATION

A UAV tracks a UGV in a global frame $\mathcal{F}_W$. The UAV carries the only exteroceptive sensor: a downward monocular camera that detects an ArUco marker on the UGV, yielding a noisy pose measurement whose quality depends on the imaging geometry. An EKF fuses these detections with UGV wheel odometry to produce a belief (μ, Σ) over the UGV state. The objective is to (i) keep the UGV inside the camera footprint so that Σ stays bounded, (ii) track the UGV robustly under bounded disturbances, and (iii) do so while limiting the number of online optimizations. The UGV must track a ground lane while remaining observable.

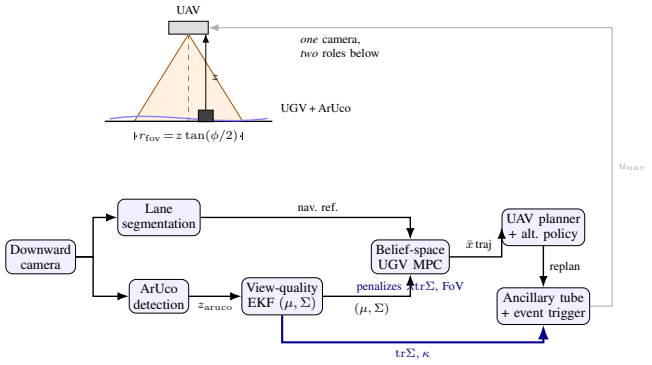


Fig. 1. System overview. *Top:* the physical geometry — a downward-camera UAV at altitude z views an ArUco-tagged UGV on a curving lane within a footprint of radius $r_{\text{fov}} = z\tan(\phi/2)$. *Bottom:* the single camera image feeds *two* roles: lane segmentation provides the UGV's navigation reference, and ArUco detection feeds the view-quality EKF (μ, Σ); both drive the belief-space UGV MPC (which penalizes $\lambda\,\text{tr}\Sigma$ and FoV exit), whose predicted trajectory the UAV planner tracks under the perception-constrained altitude policy, executed by the ancillary tube/event-trigger controller (**blue:** the EKF covariance and curvature drive the event trigger). This dual role of one sensor is the coverage–resolution coupling studied in the paper.

IV. SYSTEM ARCHITECTURE AND MODELS

Fig. 1 shows the implemented system and its signal flow.

A. UGV dynamics

The UGV uses unicycle kinematics with first-order actuator lag. Let $\mathbf{x}_g = [x_g, y_g, \theta, v, \omega]^T$ be the state, (x_g, y_g) the position, θ the heading, (v, ω) the linear/angular velocities, and $\mathbf{u}_g = [v_c, \omega_c]^T$ the commanded velocities. The discrete-time dynamics are

$$\mathbf{x}_{g,k+1} = \mathbf{f}_g(\mathbf{x}_{g,k}, \mathbf{u}_{g,k}) + \mathbf{w}_g, \tag{1}$$

where the actuator lag is governed by time constants τ_v, τ_ω and $\mathbf{w}_g$ is process noise.

B. UAV model

Although the platform is a quadrotor, the planner and the ancillary tracker treat it as a **velocity-controlled point mass** (single integrator); an inner-loop attitude/thrust controller, provided by the Gazebo velocity-control plugin, is assumed to realize commanded velocities. With position $\mathbf{p} = [p_x, p_y, p_z]^T$ and velocity command $\mathbf{u} = [v_x, v_y, v_z]^T$,

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta t\,\mathbf{u}_k + \mathbf{w}_k, \qquad A_d = I_3,\ B_d = \Delta t\, I_3, \tag{2}$$

with bounded disturbance $\mathbf{w}_k \in \mathbb{W}$. All three axes are fully actuated: v_z is a genuine decision variable and the altitude policy of Sec. IV-D is realized closed-loop in SITL (the UAV altitude spans $[2.5, 7]$ m; Sec. VII). A second-order double-integrator (acceleration input) is the more faithful rigid-body abstraction; we use (2) because it is exactly what the implementation optimizes.

C. View-quality-dependent perception

The camera's vantage point affects estimation accuracy. We use a differentiable view-quality metric $q_k \in (0, 1]$ that degrades as the UGV nears the FoV edge and as the altitude departs from a precision-optimal value z_{opt} (too low loses footprint; too high loses pixel resolution). With horizontal UAV–UGV distance $d = \|\mathbf{p}^{xy} - \mathbf{p}_g\|$ and footprint radius $r_{\text{fov}}(z) = z\tan(\phi/2)$,

$$q_k = \underbrace{\frac{1}{1 + e^{\beta(d - r_{\text{fov}})}}}_{\text{FoV gate}} \cdot \underbrace{e^{-d^2/(2\sigma_{xy}^2)}}_{\text{centering}} \cdot \underbrace{e^{-(z - z_{\text{opt}})^2/(2\sigma_z^2)}}_{\text{altitude precision}}, \quad (3)$$

with $\sigma_{xy} = r_{\text{fov}}/2$. The altitude factor makes q_k *non-monotone* in z (it does not tend to 1 as $z \to \infty$), removing the incentive to climb without bound.

The *measurement noise* follows the physically-grounded *ground-sampling-distance* (GSD) model: pixel-level localization is fixed while the GSD (m/px) grows linearly with altitude, so the metric position-measurement covariance grows *quadratically*,

$$R(z) = R_{\text{full}}\,(z/z_0)^2\,\big(1 + (d/r_{\text{fov}})^2\big), \quad (4)$$

with a mild FoV-edge inflation $(1 + (d/r_{\text{fov}})^2)$. Monotone in z, (4) captures the $R \propto z^2$ resolution side of the coverage–resolution tension (Sec. IV-D) and is the single measurement-noise model used in *all* reported results (including the fixed-altitude co-design). The view-quality metric q_k (3) is retained only as a differentiable geometric visibility gate (FoV membership and diagnostics), not as the measurement-noise model.

D. Segmentation coverage model

The same downward camera also supplies the UGV's *navigation* reference: an OpenCV segmenter extracts the lane and seeds a waypoint a lookahead distance ahead of the vehicle. That waypoint must lie inside the camera footprint or segmentation fails and the UGV loses its reference. With footprint radius $r_{\text{fov}}(z) = z\tan(\phi/2)$ and required lookahead $d_{\text{look}}(\kappa)$ — which grows with local path curvature κ, since a curving lane exits the footprint sooner and needs more forward view — define the non-dimensional *coverage factor*

$$\eta \;=\; \frac{r_{\text{fov}}(z)}{d_{\text{look}}(\kappa)} \;=\; \frac{z\tan(\phi/2)}{d_{\text{look}}(\kappa)}, \quad (5)$$

with segmentation feasible for $\eta \gtrsim 1$. This is the crux coupling: the estimator wants *low* altitude ($R \propto z^2$, better resolution), while (5) demands *sufficient* altitude for navigation. The estimation-optimal altitude is thus the lowest that keeps $\eta \gtrsim 1$, and this floor rises with curvature — the analytic basis for climbing into turns. The aerial planner (Sec. V-B) enforces $\eta \gtrsim 1$ as a soft lower bound; Sec. VII reports the empirical cost tradeoff over η.

V. Coupled Event-Triggered Tube MPC

The estimation uncertainty Σ_k enters the control formulations of both vehicles.

A. Information-aware UGV control (belief-space NMPC)

The UGV controller augments the planning state with the covariance, $\xi_k = (\mu_k, \Sigma_k)$, and *propagates both through the horizon*. At each predicted step it forms the no-observation prediction $\Sigma_{\text{pred}} = F\Sigma F^T + Q$ and a post-measurement covariance Σ_{obs} (Joseph update), and blends them by a smooth visibility gate $\gamma = 1/(1 + e^{\beta(d - r_{\text{fov}})})$:

$$\Sigma_k = (1 - \gamma)\,\Sigma_{\text{pred}} + \gamma\,\Sigma_{\text{obs}}. \quad (6)$$

The cost minimizes tracking error while penalizing predicted uncertainty and FoV exit,

$$J_g = \sum_{k=0}^{N_g - 1} \Big(\|\mathbf{e}_{g,k}\|_Q^2 + \|\mathbf{u}_{g,k}\|_R^2 + \lambda\ \text{tr}(\Sigma_k) + w_{\text{fov}}\, c_{\text{fov}}(\mathbf{x}_{g,k}) \Big), \quad (7)$$

where the FoV term is a one-sided quadratic hinge $c_{\text{fov}} = \big(\max(0,\, d_k - r_{\text{fov}})\big)^2$ (a soft constraint, not an asymptotic barrier). Crucially, the belief rollout (6) uses the *same* measurement-noise model as the on-board EKF — the GSD model (4) in all reported runs (both share one implementation, selected by a single flag) — so the planner propagates the estimator's actual uncertainty, not a surrogate. The covariance rollout (6) makes (7) a nonconvex NLP, solved with SLSQP over $N_g = 5$. The two terms $\lambda\,\text{tr}(\Sigma_k)$ and $w_{\text{fov}} c_{\text{fov}}$ are what make the controller belief-coupled and are absent from the passive baseline.

B. Aerial nominal planner and altitude policy

The UAV computes a nominal trajectory $\bar{\mathbf{p}}$ over $N_a = 10$ by optimizing goal tracking, an altitude regularizer toward a setpoint $z_k^\star$, and a visibility term that tolerates lateral offset only when altitude is sufficient:

$$J_a = \sum_k w_{\ell,k} \Big[\|\bar{\mathbf{p}}_k - \mathbf{g}\|^2 + w_z (z_k - z_k^\star)^2 + w_{\text{alt}} \frac{\|\bar{\mathbf{p}}_k^{xy} - \mathbf{g}^{xy}\|^2}{z_k^2 + 0.1} \Big] + w_u \|\bar{\mathbf{u}}_k\|^2, \quad (8)$$

where $\mathbf{g}$ is the UGV's predicted end-of-horizon position (read from the belief-MPC rollout, one direction of the coupling) and $w_{\ell,k}$ up-weights later steps. The perception-constrained altitude policy of Sec. IV-D sets this setpoint explicitly to the coverage floor plus an anticipatory margin,

$$z_k^\star \;=\; \text{clip}\big(\hat{d}_k / \tan(\phi/2) + \beta_\kappa\, \hat{\kappa}_k + z_{\text{m}},\ z_{\min},\ z_{\max}\big), \quad (9)$$

where $\hat{d}_k$ is the estimated horizontal UAV–UGV offset (the first term keeps the target within the footprint), $\hat{\kappa}_k = \max_i |\Delta\theta_i|/\Delta s_i$ is the anticipated path curvature (max discrete heading-change per arc-length over the next four predicted UGV segments[1]), and $\beta_\kappa \hat{\kappa}_k$ raises altitude into turns where the required lookahead $d_{\text{look}}(\kappa)$ — and hence the coverage floor $\eta \gtrsim 1$ (5) — grows; z_{m} is a fixed margin that keeps the forward lookahead in view on straights, and the setpoint is clipped to $[z_{\min}, z_{\max}]$. The anticipatory

[1]The curvature trigger (§V-E) uses the same discrete estimator evaluated over the first two predicted segments; the altitude policy uses up to four — same form, different horizon.

gain β_κ defaults to 0 (purely reactive to $\hat{d}_k$); the reported active-altitude runs use β_κ=5.0. In the fixed-altitude baseline $z_k^\star$=z_{opt}; either way $z_k^\star$ enters (8) as a *soft* quadratic target (weight w_z), so the planner tracks it subject to the goal and visibility terms rather than enforcing it exactly. "Climb into turns to widen the footprint" is thus a derived consequence of the coverage constraint, not a hand-coded rule.

C. Offline robust-invariant-set baseline

The ancillary law is $\mathbf{u}_k = \bar{\mathbf{u}}_k - K\mathbf{e}_k$ with tracking error $\mathbf{e}_k = \mathbf{p}_k - \bar{\mathbf{p}}_k$; substituting into (2),

$$\mathbf{e}_{k+1} = \underbrace{(A_d - B_d K)}_{=:A_K}\mathbf{e}_k + \mathbf{w}_k, \qquad \mathbf{w}_k \in \mathbb{W}. \tag{10}$$

For A_K Schur and $\mathbb{W}$ compact, the minimal RPI set is outer-approximated *offline* by a truncated Minkowski series and the constraints tightened by the Pontryagin difference [2],

$$\Omega = \bigoplus_{i=0}^{N} A_K^i \mathbb{W} \supseteq \Omega_\infty, \quad \bar{\mathbb{X}} = \mathbb{X} \ominus \Omega, \quad \bar{\mathbb{U}} = \mathbb{U} \ominus K\Omega. \tag{11}$$

This offline synthesis provides the design-time robustness budget Ω; the guarantee it yields and the lightweight online surrogate that tracks it are developed together in §VI.

D. Online adaptive ellipsoidal tube

Propagating the polytope Ω online — vertex/support-function updates and a Pontryagin difference at every 20 Hz ancillary step — is impractical on the UAV's embedded companion processor. Rather than pay that overhead, the ancillary controller uses an *ellipsoidal* invariant-set surrogate

$$\mathcal{E}(\alpha_k) = \{\mathbf{e} : \mathbf{e}^T P_0\, \mathbf{e} \le \alpha_k^2\}, \tag{12}$$

with an isotropic shape matrix $P_0 = 0.186\, I_3$ and a radius that adapts to the EKF horizontal covariance trace $\operatorname{tr}\Sigma_{xy} = \Sigma_{xx} + \Sigma_{yy}$:

$$\alpha_k = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})\,\operatorname{clip}\!\Big(\tfrac{\operatorname{tr}\Sigma_{xy} - \operatorname{tr}_{\min}}{\operatorname{tr}_{\max} - \operatorname{tr}_{\min}}, 0, 1\Big), \tag{13}$$

($\alpha_{\min} = 0.5,\ \alpha_{\max} = 2$). The tube *breathes*: it tightens when the estimate is sharp and loosens when perception degrades. We are precise about the role of (12): it is a lightweight run-time *monitor* that gates the event trigger, not an online-enforced invariant set. The certified object is the offline mRPI Ω (11) with the A_K-Schur ISS bound (§VI) — a design-time robustness budget that the online loop tracks through the monitor and the ancillary law, so tuning P_0 and $\alpha_{\min}$ sets trigger sensitivity, not invariance. A homothetic tube re-derived online is a natural extension.

E. Perception-driven event trigger

The ancillary node runs at 20 Hz but requests a nominal replan only when one of four prioritized conditions fires (implemented literally as follows).

- **(1) Tube exit (critical, no lockout):** fire iff $\mathbf{e}_k^T P_0 \mathbf{e}_k > \alpha_k^2$, i.e. the state left the adaptive ellipsoid (12).
- **(2) Covariance spike:** $\operatorname{tr}\Sigma_{xy}(t) > \tau_{\text{cov}}$ and a sudden ($>1.5\times$) rise over ~ 0.5 s (e.g. ArUco occlusion).
- **(3) Curvature anticipation:** the anticipated path curvature $\hat{\kappa}$ (Eq. 9, the wrapped-heading-change estimator over the predicted waypoints) exceeds $\tau_\kappa = 0.4$ rad/m, so the UAV climbs before the turn (raising the coverage floor of Sec. IV-D).
- **(4) Temporal drift:** elevated $\operatorname{tr}\Sigma_{xy}$ with time since last replan exceeding $T_{\min}$.

Conditions (2)–(4) carry a 1 s time-lockout (the chattering guard); condition (1) has none. Between triggers the ancillary law regulates the error via (10), and the previous nominal sequence is shifted forward. In the reported regime the perception conditions (2)–(3) dominate replanning and the tube exit acts as a rarely-binding safety backstop: the tube bounds worst-case error offline, the perception conditions schedule replanning online.

F. Nominal optimization actually solved

At each trigger the planner solves (8) over the single integrator (2) with velocity bounds $v_x, v_y \in [-1.5, 1.5]$, $v_z \in [-1, 1]$ m/s, warm-started from the previous solution (SLSQP, maxiter = 15). Altitude limits $z \in [z_{\min}, z_{\max}]$ are shaped by the cost; a hard terminal set $\mathbb{X}_f$ and the DARE terminal cost of the idealized formulation below are *not* enforced online.

VI. From Robust-MPC Synthesis to the Online Controller

The online controller is the endpoint of a deliberate design pipeline: an offline robust-MPC synthesis provides the tube and stability structure, and a lightweight online law tracks it within a companion computer's compute budget.

Design-time recursive feasibility. With the offline mRPI Ω (11) and its tightened constraints $\bar{\mathbb{X}}, \bar{\mathbb{U}}$, a nominal MPC with a control-invariant terminal set and DARE terminal cost is recursively feasible, and $\mathbf{x}_k \in \bar{\mathbb{X}} \oplus \Omega \subseteq \mathbb{X}$ [1], [2]. The online controller trades the terminal-set certificate for the millisecond-solvable ellipsoidal monitor (12) — an intentional exchange of an offline guarantee for online affordability.

Bounded inter-trigger error (ISS). With A_K Schur and $\mathbf{w} \in \mathbb{W}$ bounded, (10) is input-to-state stable. For the ancillary gain $K = \operatorname{diag}(0.6, 0.6, 0.8)$ at Δt=0.05 s the discrete error map $A_K = I - \Delta t\, K = \operatorname{diag}(0.97, 0.97, 0.96)$ is Schur ($\rho = 0.97$), giving $\limsup_k \|\mathbf{e}_k\| \le \sup\|\mathbf{w}\|/(1-\rho)$; with centimetre-scale disturbance the bound is sub-metre, so the ellipsoid membership is a meaningful monitor. The belief cost's $\lambda \operatorname{tr}\Sigma_k$ and c_{fov} terms (7) render the UGV stage cost nonconvex, so its stability is established empirically here (an information-weighted terminal cost is the natural formal route); the mRPI boxes the EKF innovation to a bounded $\mathbb{W}$, and a fully stochastic tube is a clean generalization.

Scope, and a design insight. Results are in simulation with a velocity-controlled UAV abstraction — the appropriate model for this high-level planning layer — with a full multicopter model the next step toward deployment. One structural property is a design insight, not a limitation: a *single* camera serves both navigation and estimation, so

TABLE I

COMPLETE SYSTEM CONFIGURATION, RELEASED WITH THE CODE FOR EXACT REPRODUCTION. A SINGLE COMPACT PARAMETER SET SPECIFIES EVERY STAGE; ALL VALUES ARE THOSE USED IN ALL REPORTED RUNS.

Symbol	Description	Value
Camera & measurement model (GSD)		
ϕ	Camera FoV (full angle)	1.047 rad (60°)
r_{fov}	Ground footprint radius	$z\tan(\phi/2)$
z_0	GSD reference altitude	3.0 m
R_{full}	Base ArUco position noise (x, y)	$2.5{\times}10^{-3}$ m^2
EKF, state $[x, y, \theta, v, \omega]$		
f_{ekf}	Prediction rate	20 Hz (Δt=0.05 s)
Q	Process-noise diag.	$[10^{-4}, 10^{-4}, 4{\cdot}10^{-4}, 10^{-2}, 2.5{\cdot}10^{-3}]$
UGV belief-space NMPC		
$N_g, \Delta t$	Horizon / step	5 / 0.2 s
w_d, w_ψ, w_r	Dist. / heading / rate weights	10, 10, 5
λ	Uncertainty ($\text{tr}\,\Sigma$) weight	1.0
$v_{\max}, \omega_{\max}$	UGV speed / yaw caps	0.22 m/s, 2.0 rad/s
UAV nominal (aerial) planner		
$N_a, \Delta t$	Horizon / step	10 / 0.2 s
$\bar{v}_{xy}, \bar{v}_z$	Velocity bounds	1.5, 1.0 m/s
w_z, w_{alt}	Altitude / visibility weights	3.0, 50
z_{opt}	Fixed / nominal altitude	6.0 m
$[z_{\min}, z_{\max}]$	Active-altitude band	[2.5, 7.0] m
z_m, β_κ	Policy margin / curvature gain	0.4 m, 5.0
Tube / ancillary monitor (20 Hz)		
K	Ancillary gain, $u{=}\bar{u}{-}Ke$	$\text{diag}(0.6, 0.6, 0.8)$
P_0	Ellipsoid monitor shape	$0.186\, I_3$
$[\alpha_{\min}, \alpha_{\max}]$	Adaptive monitor radius	[0.5, 2.0]
$[\text{tr}_{\min}, \text{tr}_{\max}]$	α-map range on $\text{tr}\,\Sigma_{xy}$	[0.005, 0.15]
Perception-driven event triggers		
–	Tube exit (critical)	$e^\top P_0 e > \alpha^2$
τ_{cov}	Covariance-spike threshold	0.12
τ_κ	Curvature threshold	0.4 rad/m
$\tau_{\text{drift}}, T_{\min}$	Drift threshold / min. interval	0.07, 3.0 s
T_{lock}	Non-critical lockout	1.0 s

the two are coupled through altitude and cannot be varied independently — precisely why altitude is the right control handle (Sec. VII). Isolating an observability-only benefit would require a second sensor to break the coupling; on a single-camera team, jointly optimizing coverage and resolution through altitude is the correct formulation.

VII. SIMULATION STUDY

Setup. The system runs in Gazebo with ROS 2: a TurtleBot3-class UGV with an ArUco marker, an X500-class UAV with a downward camera, an OpenCV lane/marker pipeline, the 5-state EKF, the belief-space UGV MPC, and the UAV planner/ancillary pair. Per-step telemetry (pose, ground truth, $\text{tr}\,\Sigma$, FoV distance, ArUco visibility, view quality, replan events, solve times) is logged by a common logger identical across configurations. Table I lists the *complete* configuration — one compact, subsystem-organized parameter set that fully specifies the perception, estimation, planning, tube-monitor, and event-trigger stages; every value is the one used in all reported runs and is released with the code.

Reproducibility (a methodological prerequisite). With the Gazebo GUI the pipeline is *not* reproducible — identical-configuration trajectories fork by ≈ 2.7 m and ArUco visibility swings 0–100%, as the rendering thread contends with the ROS 2 executor and perturbs the closed loop. Headless (server-only) execution removes this: trajectories agree to ≈ 0.03 m at 100% visibility. All results below are headless — an often-unstated prerequisite for reproducible closed-loop SITL evaluation.

Co-design: event-triggering makes the aerial planner affordable. The belief-space UGV controller runs continuously (it is the ground vehicle's feedback loop), so the compute lever is the *UAV nominal planner* that re-optimizes the aerial tracking trajectory. Continuous aerial replanning (C1) spends a large, sustained CPU budget; the perception-driven event trigger (C2) re-solves it only when a covariance/curvature/drift condition fires. On the nominal track (headless, R=3), C2 reduces aerial-planner invocations $\approx 25\times$ ($797 \pm 16 \to 32 \pm 1$) and its CPU $\approx 12\times$ ($3123 \pm 56 \to 253 \pm 6$ ms per 45 s run; Table II), at matched estimation quality (steady-state $\text{tr}\,\Sigma = 0.0095$ and $\approx 99.9\%$ ArUco visibility for both). Per-solve time is slightly higher for C2 (7.9 ± 0.4 vs. 3.9 ± 0.1 ms, a warm-start effect), but the $\approx 25\times$ solve-count reduction dominates the net saving, making perception-aware aerial replanning affordable on a constrained companion computer at estimation quality indistinguishable from continuous replanning. The one cost is a small loss of inter-trigger smoothness (the held nominal raises high-frequency content), a favorable tradeoff. Isolating the *perception-specific* conditions against a plain error trigger is future work.

TABLE II

AERIAL-PLANNER CO-DESIGN METRICS, FIXED SYSTEM (HEADLESS, NOMINAL TRACK, MEAN ± STD OVER R=3). C1 = CONTINUOUS AERIAL REPLANNING; C2 = PERCEPTION-DRIVEN EVENT-TRIGGERED. THE BELIEF-SPACE UGV CONTROLLER RUNS CONTINUOUSLY IN BOTH. STEADY-STATE WINDOW $t \geq 8$ S. ESTIMATION QUALITY IS MATCHED; THE COMPUTE BUDGET IS NOT.

Metric	C1 (Continuous)	C2 (Event-Trig.)
Aerial-planner solves (# / run)	797 ± 16	32 ± 1
Mean solve time (ms)	3.9 ± 0.1	7.9 ± 0.4
Total aerial-planner CPU (ms / run)	3123 ± 56	253 ± 6
Steady-state $\text{tr}\,\Sigma$	0.0095	0.0095
ArUco visibility (%)	99.8	99.9
UAV–UGV offset (m)	0.128 ± 0.002	0.060 ± 0.005

The coverage–resolution altitude tension. With the GSD model enabled, we sweep altitude across three tracks of increasing curvature (headless; low-altitude cells are the median of R=3 reps). Two costs move oppositely with altitude (Fig. 2): estimation uncertainty $\text{tr}\,\Sigma$ *decreases* as altitude decreases (0.0094 at z=6 m to 0.0053 at z=2 m — the $R \propto z^2$ resolution gain, and track-independent), while the UGV steering effort needed to keep the path segmented *increases* as altitude decreases, and increases with path curvature. This is the tension the altitude policy must resolve: fly as low as segmentation allows, and climb where the navigation floor rises.

The adaptive policy dominates fixed altitude. We close the loop with the perception-constrained policy and compare

it against the two natural fixed-altitude baselines on the hairpin (the most demanding track; headless, GSD noise, R=3). The policy actuates altitude over $[2.5, 5.1]$ m — holding near the low-altitude floor on straights and climbing through turns (Fig. 3) — and it is *not* matched by any static choice (Fig. 3). The **fixed-low** baseline (z=2.5 m) obtains good average resolution but suffers coverage-loss transients: as the target leaves the small footprint in the hairpin, its peak covariance spikes to 0.14 (9× the adaptive policy's peak of 0.016) and it drops marker visibility. The **fixed-high** baseline (z=7 m) never loses coverage but pays a 61% estimation penalty ($\mathrm{tr}\,\Sigma = 0.0105$ vs. 0.0065) from the degraded GSD. The **adaptive** policy attains the low mean covariance of flying low (0.0065, matching the fixed-low 0.0062) while keeping the peak covariance and FoV exits of flying high — the best of both jaws of the tension, with 100% marker visibility. We are precise about one point: because the horizon is short and closed-loop tracking adds lag, the achieved altitude is roughly coincident with, rather than cleanly leading, each turn; the strict anticipation property (Prop. 3) is established analytically and in the exact numerical model. The SITL claim we make is the one the data supports and that matters operationally — an adaptive altitude that dominates every fixed choice.

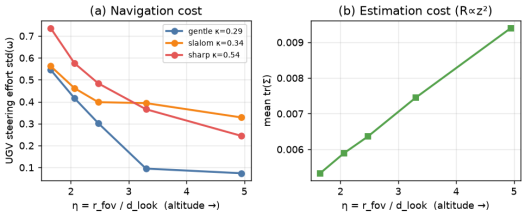


Fig. 2. The two jaws of the altitude tension ($\eta = r_{\mathrm{fov}}/d_{\mathrm{look}}$, increasing with altitude). *Left:* UGV steering effort (navigation cost) rises at low altitude and with path curvature. *Right:* estimation uncertainty $\mathrm{tr}\,\Sigma$ falls at low altitude (GSD, $R \propto z^2$; track-independent). The optimum flies as low as segmentation allows.

Consistent with the shared-camera coupling (§VI), $\mathrm{tr}\,\Sigma$ at a given altitude is dominated by the shared measurement noise and is nearly track-independent, which is why coverage and resolution are traded jointly through altitude. One regime resists a crisp boundary: at z=2 m and UGV speed 1.0 m/s — where the coverage constraint binds hardest — residual simulator timing sensitivity makes marker visibility vary run-to-run, so we characterize the tension by the envelope and the dominance result above rather than by a single feasibility contour there.

VIII. CONCLUSIONS

We studied aerial–ground cooperative tracking where a single downward UAV camera is the ground vehicle's sole exteroceptive sensor, serving both its navigation reference and its state estimate. We identified the coverage–resolution altitude tension intrinsic to this shared-sensor architecture and demonstrated closed-loop that the perception-constrained altitude policy resolving it dominates both fixed-altitude

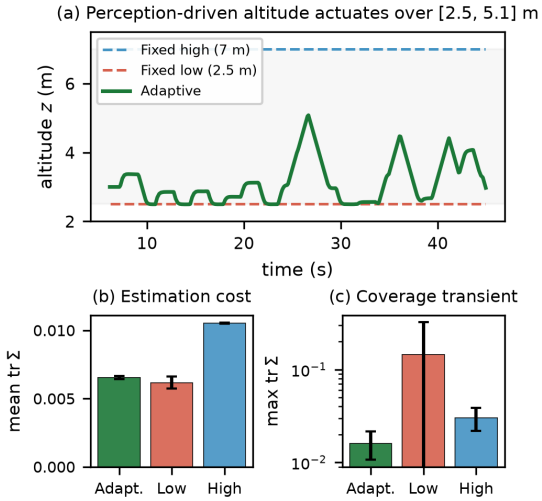


Fig. 3. Adaptive altitude dominates fixed altitude (hairpin, headless, GSD noise, R=3). *(a)* The policy actuates altitude over $[2.5, 5.1]$ m, holding near the low-altitude floor on straights and climbing through turns, versus the two static baselines. *(b)* A fixed-high policy pays a 61% mean-estimation penalty. *(c)* A fixed-low policy suffers coverage-loss transients (9× larger peak $\mathrm{tr}\,\Sigma$, log scale). The adaptive policy is best on both axes.

baselines — avoiding the fixed-low policy's coverage-loss transients (9× smaller peak covariance) and the fixed-high policy's 61% estimation penalty. We further showed a perception-driven event trigger cuts the UAV nominal (aerial) planner's re-solves ≈ 25× and its CPU ≈ 12× at matched estimation quality, and we release the reproducible testbed. Future work: a full multicopter model for hardware-faithful altitude dynamics; a second exteroceptive sensor to isolate the observability benefit; and a homothetic tube carrying recursive-feasibility guarantees into the online loop.

REFERENCES

[1] D. Q. Mayne, M. M. Seron, and S. V. Raković, "Robust model predictive control of constrained linear systems with bounded disturbances," *Automatica*, vol. 41, no. 2, pp. 219–224, 2005.

[2] S. V. Rakovic, E. C. Kerrigan, K. I. Kouramas, and D. Q. Mayne, "Invariant approximations of the minimal robust positively invariant set," *IEEE Trans. Autom. Control*, vol. 50, no. 3, pp. 406–410, 2005.

[3] D. Q. Mayne, J. B. Rawlings, C. V. Rao, and P. O. M. Scokaert, "Constrained model predictive control: Stability and optimality," *Automatica*, vol. 36, no. 6, pp. 789–814, 2000.

[4] D. Falanga, P. Foehn, P. Lu, and D. Scaramuzza, "PAMPC: Perception-aware model predictive control for quadrotors," in *IEEE/RSJ Int. Conf. Intelligent Robots and Systems (IROS)*, 2018.

[5] R. Platt, R. Tedrake, L. Kaelbling, and T. Lozano-Pérez, "Belief space planning assuming maximum likelihood observations," in *Robotics: Science and Systems (RSS)*, 2010.

[6] J. van den Berg, S. Patil, and R. Alterovitz, "Motion planning under uncertainty using iterative local optimization in belief space," *Int. J. Robotics Research*, vol. 31, no. 11, pp. 1263–1278, 2012.

[7] A. Eqtami, D. V. Dimarogonas, and K. J. Kyriakopoulos, "Event-triggered control for discrete-time systems," in *American Control Conference (ACC)*, 2010, pp. 4719–4724.

[8] W. Langson, I. Chryssochoos, S. V. Raković, and D. Q. Mayne, "Robust model predictive control using tubes," *Automatica*, vol. 40, no. 1, pp. 125–133, 2004.

[9] D. Q. Mayne, E. C. Kerrigan, E. J. van Wyk, and P. Falugi, "Tube-based robust nonlinear model predictive control," *Int. J. Robust Nonlinear Control*, vol. 21, no. 11, pp. 1341–1353, 2011.

[10] L. Chisci, J. A. Rossiter, and G. Zappa, "Systems with persistent disturbances: Predictive control with restricted constraints," *Automatica*, vol. 37, no. 7, pp. 1019–1028, 2001.
[11] W. P. M. H. Heemels, K. H. Johansson, and P. Tabuada, "An introduction to event-triggered and self-triggered control," in *IEEE Conf. Decision and Control (CDC)*, 2012, pp. 3270–3285.
[12] P. Tabuada, "Event-triggered real-time scheduling of stabilizing control tasks," *IEEE Trans. Autom. Control*, vol. 52, no. 9, pp. 1680–1685, 2007.
[13] H. Li and Y. Shi, "Event-triggered robust model predictive control of continuous-time nonlinear systems," *Automatica*, vol. 50, no. 5, pp. 1507–1513, 2014.
[14] I. Spasojevic, V. Murali, and S. Karaman, "Perception-aware time optimal path parameterization for quadrotors," in *IEEE Int. Conf. Robotics and Automation (ICRA)*, 2020, pp. 3213–3219.
[15] Z. Zhang and D. Scaramuzza, "Beyond point clouds: Fisher information field for active visual localization," in *IEEE Int. Conf. Robotics and Automation (ICRA)*, 2019, pp. 5986–5992.
[16] S. Prentice and N. Roy, "The belief roadmap: Efficient planning in belief space by factoring the covariance," *Int. J. Robotics Research*, vol. 28, no. 11–12, pp. 1448–1465, 2009.
[17] V. Indelman, L. Carlone, and F. Dellaert, "Planning in the continuous domain: A generalized belief space approach for autonomous navigation in unknown environments," *Int. J. Robotics Research*, vol. 34, no. 7, pp. 849–882, 2015.
[18] J. A. Placed, J. Strader, H. Carrillo, N. Atanasov, V. Indelman, L. Carlone, and J. A. Castellanos, "A survey on active simultaneous localization and mapping: State of the art and new frontiers," *IEEE Trans. Robotics*, vol. 39, no. 3, pp. 1686–1705, 2023.
[19] C. Cadena, L. Carlone, H. Carrillo, Y. Latif, D. Scaramuzza, J. Neira, I. Reid, and J. J. Leonard, "Past, present, and future of simultaneous localization and mapping: Toward the robust-perception age," *IEEE Trans. Robotics*, vol. 32, no. 6, pp. 1309–1332, 2016.
[20] Z. Li, R. Mao, N. Chen, C. Xu, F. Gao, and Y. Cao, "ColAG: A collaborative air-ground framework for perception-limited UGVs' navigation," in *IEEE Int. Conf. Robotics and Automation (ICRA)*, 2024, pp. 16781–16787.
[21] M. Ferreira Santos, P. Castillo, and A. Corrêa Victorino, "Aerial vision based guidance and control for perception-less ground vehicle," in *European Control Conference (ECC)*, 2024, pp. 2780–2785.
[22] B. Grocholsky, J. Keller, V. Kumar, and G. Pappas, "Cooperative air and ground surveillance," *IEEE Robotics Autom. Mag.*, vol. 13, no. 3, pp. 16–25, 2006.
[23] J. Thomas, J. Welde, G. Loianno, K. Daniilidis, and V. Kumar, "Autonomous flight for detection, localization, and tracking of moving targets with a small quadrotor," *IEEE Robotics Autom. Lett.*, vol. 2, no. 3, pp. 1762–1769, 2017.
[24] B. Penin, P. Robuffo Giordano, and F. Chaumette, "Vision-based reactive planning for aggressive target tracking while avoiding collisions and occlusions," *IEEE Robotics Autom. Lett.*, vol. 3, no. 4, pp. 3725–3732, 2018.
[25] S. I. Roumeliotis and G. A. Bekey, "Distributed multirobot localization," *IEEE Trans. Robotics Autom.*, vol. 18, no. 5, pp. 781–795, 2002.
[26] S. S. Kia, S. Rounds, and S. Martínez, "Cooperative localization for mobile agents: A recursive decentralized algorithm based on Kalman-filter decoupling," *IEEE Control Syst. Mag.*, vol. 36, no. 2, pp. 86–101, 2016.